INCH-POUND

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30 March 1992
SUPERSEDING
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MILITARY SPECIFICATION

BORING MACHINE, CAM AND MAIN BEARING, PORTABLE

This specification is approved for use by all Department and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type of portable, universal boring machine for line boring, facing, and filleting cam and main bearings in motor vehicle engine blocks.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-corner.

MILITARY

MIL-P-116	-	Preservation, Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910

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STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Service.
FED-STD-101	-	Test Procedures For Packaging

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robins Avenue, Philadelphia, PA 19111-5094)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGISTER (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 2000-A - Gear Classification and Inspection Handbook.
AGMA 390.03A - Gear Handbook, Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 N. Fort Myer Drive, Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI IEEE 268 - Metric Practice.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E380 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches.
ISO 1122/1 - Glossary of Gear Terms - Part 1: Geometrical Definitions.

(Applications for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 First article. When specified (see 6.2.1), the contractor shall furnish one or more boring machines for first article inspection (see 4.3 and 6.2.1). The first article may be either a preproduction model or an initial production item(s) which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The boring machine shall be new and of the manufacturer's latest design which shall meet the performance requirements specified herein. The machine shall be an engine block attachment type that is portable and universal. It shall be designed for line boring, facing, and filleting cam and main bearings in motor vehicle engine blocks. The boring bar shall be capable of being driven from either end by an electric drill, not furnished with the machine, or by a handcrank. The design shall be such that all parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the machine or any machine component.

3.2.1 Measurement system. The US Customary Systems of Units (US) or the International System of Units (SI) may be used in the design and construction of the boring machine. In this specification, all measurements, dimensions, sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380 or ANSI IEEE 268, as applicable.

3.2.2 Measuring and indicating devices. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI systems of measurement. When a dual US and SI system is furnished, feed dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system (see 3.6.7).

3.2.3 Interchangeability. All parts of the machine shall be manufactured to definite standards and tolerances. Parts shall be interchangeable without requiring modification of the part or its mating components. Parts subject to possible future replacement as spare or repair parts shall be assigned a manufacturer's part number and shall be identified in the repair parts manual (see 6.3.1). All replacement, repair, and spare parts shall provide the same or better form, fit, function, and performance as the parts initially installed in the boring machine. When industry or Government standards are applicable, replacement parts shall conform to those standards.

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3.2.4 Lubrication. All bearings (except sealed-for-life lubricated type), mating gears, and all other moving parts shall be provided with means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the machine tool industry (see 6.5). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the machine to remove the temporary caps and stoppers before start-up or use of the boring machine.

3.2.5 Gears. All gears and pinion designs, selected for use in the boring machine or its components shall be manufactured to either English (USA) standard inch diametral pitch per AGMA 390.03A and AGMA 2000-A, or Metric (SI) module ratio per the International Organization for Standardization (ISO) ISO 54 and ISO 1122/1.

3.2.6 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.7 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. Guards shall be removable to provide access to guarded parts. The boring machine shall comply with the general safety and health requirements promulgated under 29 CFR 1910 that are applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry, shall be suitable for the intended purpose of the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

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3.4 Construction. The boring machine shall be constructed to be capable of performing the operations specified herein. The machine shall be complete so that, when installed and connected to the proper power source, the machine may be used for all operations for which it was designed.

3.4.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness or adjustment. The methods for preventing loss of tightness or adjustment shall be in accordance with accepted engineering standards and practice. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.4.4 Surfaces. Castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.4.5 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.5 Performance and product characteristics.

3.5.1 Capacity. The boring machine shall perform the operations specified herein, within a bearing diameter range of 1.5 through 4 inches, in engine blocks up to, and including, 60 inches in length, 20 inches in width, and 11 inches in height to bearing center line from the crankcase pan rail surface. The boring machine shall be mounted on the crankcase pan rail surface and shall be capable of being transferred as a unit from one engine block to another for line boring.

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3.5.2 Feed range. A minimum feed range of 0.0035 to 0.0070 inch per revolution of the boring bar shall be provided by the feed mechanism..

3.5.3 Accuracy. The boring machine shall be capable of producing finished bearing surfaces that are concentric and without taper to within ±0.0005 inch for each bearing surface and having a surface roughness no greater than 24 microinches arithmetical average (aa). The boring machine shall be capable of line boring the bearing surfaces over the maximum length of the bar, in line, to within 0.003 inch.

3.5.4 Weight. The total weight of the boring machine, including equipment and case, shall not exceed 300 pounds (see 3.6.8).

3.6 Components. The boring machine shall consist of, but not be limited to, the principal components specified in 3.6.1 thru 3.6.8.

3.6.1 Mounting devices. Mounting devices shall be provided to anchor the boring machine securely to the engine block. The devices shall incorporate means for accurate positioning of the boring bar relative to the mean center of the cam and main bearings. All tools and fixtures required for the setup or operation of the boring machine on engine blocks within the specified capacity shall be provided with each machine.

3.6.2 Boring bar. The boring bar shall have internal tool bit holding provisions located along the bar that permit adjustment of the tool bit at the bearing to be finished. Eight lock screws for securing tool bits in boring bars shall be furnished with each boring machine.

3.6.3 Tool bits. No less than four high speed tool bits shall be provided with each machine for each boring, facing, and filleting operation. The tool bits shall be ground for their intended use and have a Rockwell hardness of 63 to 65 on the C-scale. Unless otherwise specified (see 6.2.1), four tool bits with carbide tips shall be furnished with each machine for the boring of cast iron bearing saddles.

3.6.4 Feed mechanism. The feed mechanism shall provide not less than two feeds covering the feed range specified in 3.5.2. the feed mechanism shall operate both manually and automatically.

3.6.5 Handcrank. A handcrank shall be provided with each machine to manually turn the boring bar (see 3.2).

3.6.6 Flexible coupling. A detachable flexible coupling shall be provided to transfer the power from a 0.5 inch electric drill, not furnished with the machine, to the boring machine (see 3.2).

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3.6.7 Measuring devices. A direct reading micrometer for measuring bearing bore diameters within a range of 1.25 to 4 inches, readable to 0.0001 inch, shall be included. All gages, special micrometers, and dial indicators required for the normal setup and operation of the boring machine shall be provided with each machine. All measuring devices furnished shall be permanently marked with clear, bold face characters that can be easily read (see 3.2.2).

3.6.8 Carrying case. The case shall be made of wood or steel designed to contain, protect, and carry the boring machine with its component parts, equipment, and tools. The carrying case shall be fitted with a hinged lid with positive acting latches. The carrying case shall have a carrying handle(s) suitable for operator handling or sling handling (see 3.5.4).

3.7 Product identification. The name-plate and the information-plate may be incorporated into one or more corrosion resistant metallic plate(s) at the manufacturer's option. The plates shall be in accordance with MIL-STD-130. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The nameplate shall be secured to the machine with permanent fasteners such as screws, bolts, and rivets at each of the four corners. The nameplate shall include the following information:

Nomenclature National Stock Number (NSN) Manufacturer's name
Manufacturer's model number Manufacturer's serial number (if applicable)
Contract number

3.8 Workmanship. Standards of workmanship shall assure that the boring machine shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the

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specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this dose not authorize submission of know defective material, ether indicated or actual, nor does it commit the government to accept of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production units being offered for acceptance under the contract. The inspection shall consist of a through e as follows and failure of any unit to pass these inspections or tests shall be cause for rejection of the unit.

(a) Product examination (4.5)
(b) Performance test (4.6.1)
(c) Accuracy test (4.6.2)
(d) Packaging validation test (4.6.3)
(e) Packaging inspection (4.7)

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or Test	Lot or Batch Size	Sample Size
Product examination (4.5)	1-90	100%
Performance test (4.6.1)	1-90	13
Accuracy test (4.6.2)	1-90	13
Packaging validation test (4.6.3)	1/	-
Packaging inspection (4.7)	1-90	3

The lots shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size, 100 percent inspection is required. Each lot shall be accepted with no defects and rejected if one or more defects are found.

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1/ The packaging validation tests shall be conducted on one of the packaged items selected from the first lot submitted for quality conformance inspection and tests.

4.5 Product examination. Visually, dimensionally, and manually examine each boring machine to determine conformance with the requirements of 3.2 through 3.4.5 and 3.5.4 through 3.8. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examination includes measuring dimensions as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacturing at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operation. Failure of the contractor to provide objective evidence that the item and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.2.3), shall be cause for rejection.

4.6 Tests

4.6.1 Performance test. The boring machine shall be assembled to a standard commercial six or eight cylinder engine block within the capacity range specified in 3.5.1. After boring one bearing with power provided by the handcrank, a 1/2 inch electric drill shall be used for the remainder of the test until all cam and main bearings in the block have been bored, faced, and filleted. During the test, the boring machine shall be examined for compliance with 3.2, 3.5.1, and 3.5.2.

4.6.2 Accuracy test. When the test specified in 4.6.1 has been completed, the inside diameter, surface finish, and in-line accuracies of the bearings shall be measured to determine compliance with 3.5.3.

4.6.3 Packaging validation tests. After the Quality Assurance inspection and tests have been successfully completed, one boring machine, packaged in accordance with the requirements of Section 5 Packaging, shall be selected at random and subjected to the following rough handling tests specified in FED-STD-101 (see 5.7). If a First Article Test (FAT) is required by the contract, the packaging validation test shall be conducted on the item used for the FAT. If a FAT is not required by the contract, then an item from the first production lot shall be used for the packaging validation tests. The apparatus used shall be as described in FED-STD-101. A dummy load shall not be used. All tests are to be conducted under ambient conditions. Environmental temperature/humidity tests are not required. After each packaging validation test has been completed, the container shall be

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inspected. Defects in accordance with those described below shall be cause for rejection. The unit shall then be unpacked and again subjected to the Quality Assurance inspections and tests to reaffirm that the Item meets the requirements. After all requirements have been met, the unit shall be inspected and repackaged (see 4.7) before submission for government acceptance. After testing, any damage to the packaging shall be replaced prior to acceptance of the lot.

4.6.3.1 Mechanical handling test. Test shall be conducted using Method 5011.1 Paragraphs 6.2 (Lifting and transportation by forklift truck), of FED-STD-101. Criteria for passing/failing of the test is stated in Paragraph 7.1.5 of Method 5011.1.

4.7 Packaging inspection. The items shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing, and marking requirements specified in Section 5 of the specification.

5. PACKAGING

5.1 Disassembly. When practical, for the protection of components, attachments and accessories against damage, pilferage or to reduce cubage, items may be disassembled only to a necessary basic degree. Disassembled parts shall be preserved, anchored, braced, blocked and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special skills or tools.

5.2 Disassembled parts marking. Disassembled parts shall be marked, when necessary with instructions, to facilitate reassembly. Removed parts and mating parts on the items shall be marked by stenciling, on the part, or by use of tags. Tags shall be waterproofed and ink used (on parts or tags) shall be waterproof. Tags shall be firmly secured to each individual part.

5.3 General requirements. Component parts of machinery, accessories, and maintenance tools shall be cleaned, dried, preserved, packaged, and packed as specified herein. Parts and assemblies which are not covered herein shall be preserved and packaged utilizing the materials and methods specified for similar parts and assemblies described herein. Cleaning and drying procedures and P-type preservatives shall conform to MIL-P-116 and the criteria and guidelines specified therein.

5.4 Preservation. Preservation shall be Level A or C as specified (see 6.2.1).

5.4.1 Level A preservation.

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5.4.1.1 Cleaning and drying. Thorough cleaning and drying of all bare ferrous metal surfaces is essential before the application of a preservative or other protective finish such as paint, plating, etc. Care must be taken to limit the handling of cleaned parts prior to the assembly of new machinery. Where practicable, processors and assemblers shall wear rubber or plastic gloves while handling cleaned parts.

5.4.1.2 Internal surfaces. All ferrous metal surfaces of parts which will be utilized internally in the assembly of new machinery shall be solvent cleaned followed by fingerprint remover and wiped or drain dried. In the interim, to preclude contamination from forming on cleaned surfaces prior to the assembly operation, parts shall be coated with a light weight preservative oil conforming to P-9. To prevent dust from accumulating on cleaned parts, cover with a sheet of chemically neutral paper.

5.4.1.3 External surfaces. All accessories and attachments such as chucks, faceplates, steady rests, toolholders, abrasive wheels, etc., shall be removed from the assembly prior to the external cleaning operation. All external, critical ferrous metal surfaces, fit or function, of the machine shall be solvent cleaned followed by fingerprint remover and wiped or drain dried. Noncritical ferrous metal surfaces shall be solvent cleaned and wiped or drain dried. Nonmetallic composites and painted metal surfaces shall be cleaned by wiping with a clean, dry cloth. Care must be taken to prevent cleaning solvents from coming in contact with nonmetallic composites. When cleaned surfaces of the machine show signs of rusting before application of the final preservative, reprocessing surface shall be confined to that area of the assembly. Alternatively, the solvent used for the final wiping operation shall contain a small amount, approximately 5 percent by volume, preservative oil conforming to P-10, type 10, grade 30. It is extremely important that elapsed time between these operations be held to an absolute minimum.

5.4.1.4 Disassembled components. Disassembled components detached from the basic machine shall be cleaned and dried as specified for like items.

5.4.1.5 Machine tools and tool accessories. All such tools and tool accessories shall be clean and free from contaminants. Preservatives shall be applied to provide protection from corrosion during shipment and storage. All preserved items shall be wrapped with greaseproof barrier material. Critical surface items, i.e., parallels, etc., shall also be wrapped with no less than 1/2 inch uncompressed cushioning material.

5.4.1.6 Electrical equipment. Electrical equipment shall be clean and dry. Preservatives shall be applied as required to prevent corrosion to moving or other ferrous metal parts, but shall not come into contact with electrical connection, windings, etc. Openings shall be sealed with cap-plugs or with waterproof, pressure sensitive tape to provide protection against entry of any foreign materials.

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5.4.1.7 Lubricating systems. Vent openings in the lubricating systems shall be temporarily capped or plugged to prevent loss of lubricants during shipment and handling. Tags, warning that "ALL TEMPORARY CAPS OR PLUGS MUST BE REMOVED BEFORE START-UP OR USE OF THE ITEM", shall be attached to all temporary caps or plugs intended to be vented to the atmosphere during operation. These warnings shall also be included in the instruction manual. In the event a lubricant reservoir cannot be covered and sealed, it shall be appropriately preserved and the lubricant shall be provided in a sealed container and shipped with each machine. A warning tag shall be placed in a conspicuous place near the filling area, warning the user to "INSTALL THE FURNISHED LUBRICANT BEFORE OPERATION OF THE EQUIPMENT". The operator's manual shall also reflect the above warning and procedure.

5.4.1.8 Nonlubricated interior surfaces. Surfaces of internal gears which do not run in lubricant shall be thoroughly coated with P-2, grade 2 preservative.

5.4.1.9 Sliding or friction surfaces. Surfaces of ways, gibs, driving gears, traversing and adjusting screws, and other critical, exterior operating machined surfaces shall be coated with P-2, grade 2 preservative compound and covered with greaseproof barrier material. Parts of the machine which are adjustable shall be locked into place.

5.4.1.10 Bearings and journals. Grease lubricated bearings shall be preserved with P-11, preservative. Oil lubricating journals shall be filled with lubricating oil P-10, type 1, grade optional.

5.4.1.11 Electrical wiring system. Exposed ends of wire, plug openings, sockets, connector plugs, terminals, and openings into switches and junction boxes, etc., shall be sealed with waterproof tape or cap-plugs, respectively. Electric cable shall be coiled to a minimum safe diameter and secure with cotton tape or twine to prevent uncoiling.

5.4.1.12 Closed dial indicators. Closed dial indicators shall be covered with 1 inch thick uncompressed cushioning material and secured in place with waterproof, pressure sensitive tape. Cushioning materials and tape used shall be water-resistant, non-dusting, and non-corrosive.

5.4.1.13 Closure of openings. Small openings, except vents and louvers installed for ventilation purposed, shall be sealed with waterproof, pressure sensitive tape. Large openings shall be covered with waterproof barrier material. Secure materials in place with tape specified above. When very large openings are covered, or when location of openings render coverings vulnerable to puncture, coverings shall be protected by applying wood, plywood, sheet metal of fiberboard over the materials. Open ends of all piping and fittings shall be plugged to prevent the entrance of foreign materials, using pipe fittings, waterproof tape or plastic cap-plugs.

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5.4.1.14 Technical data. Technical data furnished with the machine shall be packaged in a waterproof, greaseproof barrier material and sealed. The package shall be legibly and clearly marked "TECHNICAL DATA", in bold letters. The package shall be protected against puncture and abrasion. The data shall then be positioned in the carrying case.

5.4.1.15 Consolidated packaging. Packaged accessories, maintenance tools, and small disassembled components shall be placed in the carrying case. When it is not feasible to stow parts as specified above, place parts in fiberboard box or boxes, class weather-resistant, of sufficient strength to handle weight inside.

5.4.1.16 Cushioning. Cushioning materials shall be water resistant, non-dusting and non-corrosive and of sufficient thickness to prevent damage, not less than 1/2 inch thick wrap.

5.4.1.17 Protection of surfaces. All exposed preserved surfaces of machines which are subject to being disturbed during the packing operation shall be loosely covered with greaseproof barrier material. Not less than two layers of similar-type barrier material shall be used to separate wood or metal blocking and finished metal surfaces of the machine. Secure barrier in place with waterproof, pressure sensitive tape.

5.4.1.18 Moveable parts. Operating heads, rams, counter-balances, and other moving parts shall be secured by mechanical devices incorporated in the machine. The devices alone do not always provide adequate support to prevent movement in the handling and transporting of the item and therefore must be augmented with appropriate wood blocking, metal tie rods used in conjunction with wood blocking, metal strapping, or other devices which will immobilize the part. Handwheels, cranks, levers, and similar controls shall be secured in position with common annealed wire or steel strapping.

5.4.1.19 Micrometers. Measurement indicating devices, micrometers, etc., shall be enclosed with no less than 1 inch thick uncompressed cushioning material and placed in the carrying case.

5.4.2 Level C. Level C preservation shall be in accordance with the requirements of MIL-STD-1190.

5.5 Packing. Packing shall be level A, B, or C as specified.

5.5.1 Level A packing. Unless otherwise specified, machinery, including accessories, weighing less than 2500 pounds, shall be packed in cleated plywood boxes conforming to PPP-B-601, overseas type; or nailed wood boxes conforming to PPP-B-621, class 2, style 2, 3, 4, or 7. Selection of containers shall be governed by the size and weight limitations of the applicable container specification. Box closure and strapping requirements

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shall be in accordance with the appendix to the applicable box specification. The arrangement of contents, anchoring, blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186.

5.5.2 Level B packing. Level B packing shall be the same as for Level A, except that the boxes shall be domestic in lieu of overseas type.

5.5.3 Level C packing. Level C packing shall be in accordance with the requirements of MIL-STD-1190.

5.5.4 Consolidated containers. Containers and items not positioned in the carrying case shall be placed inside the shipping container housing the machine. The boxes shall be restrained by use of metal strapping or tie rod assemblies.

5.6 Marking. Marking shall be in accordance with MIL-STD-129. Bar code marking is also required and shall be in accordance with MIL-STD-129.

5.7 Rough handling. The completed unit pack, shipping container(s) and all contents shall be capable of withstanding the shipping and handling conditions expected from fork lift handling and truck transport. The container, its contents and the packaging shall not be damaged, functionally or physically, or items displaced by the appropriate rough handling test(s) specified elsewhere in the contract. Tests stated in Section 4 Quality Engineering Requirements shall be performed.

5.8 Workmanship. Workmanship shall be such that when proper procedure is followed, materials and equipment being processed will be provided the maximum protection against corrosion and deterioration, consistent with the shipment hazards known or anticipated, and suitable for long term storage.

6. NOTES

6.1 Intended use. This specification covers a universal boring machine that is portable and used to line bore, face, and fillet cam and main bearings in motor vehicle engine blocks. The machine is powered by either a hand operated crank that is furnished with the machine or by a 1/2 inch (or larger) electric drill that is not furnished.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Measuring and indicating device graduation(s) required (see 3.2.2).

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(d) Exceptions and additional safety/health requirements when required (see 3.2.7 and 6.4).
(e) Painting and finishing if different (see 3.4.5).
(f) Carbide tool bits if different (see 3.6.3).
(g) First article inspection if different (see 4.3).
(h) Quality conformance inspection if different (see 4.4).
(i) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3.1 Repair part manuals. Repair part and accessory identification, in repair part manuals, shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer(, Commercial and Government Entity (CAGE), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298. (see 3.2.3).

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Standard lubricants. The machine designer should specify and use, to the maximum extent possible, those lubricants (see 3.2.4) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 can be used as selection guides.

6.6 Subject term (key word) listing.

line boring
facing
filleting
Boring Machine, Cam

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:
Army - AL
Air Force - 99

Review activity:
Air Force - 82

User activity:
Navy - MC

Preparing activity:

Army - AL

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